A participation counterexample and a payment repair

For the Q7 resource-allocation game

4 November 2026

Scope. Garjani, Shokri and Kebriaei [1] propose a quadratic-payment mechanism and claim efficient Nash implementation, unique equilibrium, budget balance, individual rationality and stochastic learning convergence. This note gives separate counterexamples to participation and uniqueness for their explicit payment, and proves a static participation repair. It does not establish convergence of a learning algorithm or priority of the repaired mechanism.

1 The game and the payment

There are $N \geq 2$ players and K resources with capacity $c \in \mathbb{R}^K_+$. Player n has a compact convex set $X_n \subset \mathbb{R}^K_+$ containing zero and a private valuation $V_n(x_n) = \mathbb{E}[\psi_n(x_n; \xi_n)]$, with $V_n(0) = 0$. The planner's problem is

$$\max_{x_n \in X_n} \sum_{n=1}^{N} V_n(x_n) \quad \text{subject to} \quad \sum_n x_n \leq c. \tag{1}$$

Player n submits a quantity and a nonnegative price vector, $s_n = (x_n, p_n) \in X_n \times \mathbb{R}^K_+$. The allocation is exactly x_n; shared capacity is not a constraint on unilateral messages. Utility is $U_n(s) = V_n(x_n) - t_n(s)$: positive tax is paid by the player, and negative tax is a subsidy. A player submits neither its valuation function nor a report of its type.

Definition 1. *A profile s^* is a Nash equilibrium if $U_n(s_n^*, s_{-n}^*) \geq U_n(s_n, s_{-n}^*)$ for every player and every admissible joint quantity–price deviation s_n. Individual rationality means $U_n(s^*) \geq V_n(0) = 0$ for every player. Budget balance means $\sum_n t_n(s^*) = 0$.*

Write the paper's payment scale as $\gamma > 0$ (its notation is α). Its explicit payment, regrouped algebraically, is

$$\begin{aligned} t_n^{\mathrm{orig}}(s) = {} & \underbrace{\frac{\gamma}{2(N-1)} \sum_{m \neq n} \|p_n - p_m\|^2}_{\text{price disagreement}} \\ & \underbrace{- \frac{\gamma}{N-1} p_n^\top \left(\sum_m x_m - c \right)}_{\text{excess-demand signal}} + \underbrace{\frac{\gamma N}{(N-1)^2} \left(\sum_{m \neq n} p_m \right)^\top (x_n - c/N)}_{\text{quantity charge}} \\ & \underbrace{- \frac{\gamma}{(N-1)^2} \sum_{m \neq n} p_m^\top (x_m - c/N)}_{\text{opponent-only adjustment}} . \end{aligned} \tag{2}$$

The first term alone makes the best price the others' average. The first two together, holding quantities fixed, give

$$p_n^{\mathrm{BR}} = \left[\frac{1}{N-1} \sum_{m \neq n} p_m + \frac{\sum_m x_m - c}{N-1} \right]_+ ,$$

where projection onto the nonnegative orthant is coordinatewise. Excess demand therefore raises proposed prices. The second and third terms together give

$$\nabla_{x_n} t_n^{\text{orig}} = \gamma \left(-\frac{p_n}{N-1} + \frac{N}{(N-1)^2} \sum_{m \neq n} p_m \right) = \gamma p \quad \text{when all } p_m = p.$$

The reference c/N changes transfer levels, not marginal quantity incentives. The last term affects the recipient's utility but not its best response, since it depends only on opponents' messages.

2 A three-player, one-resource counterexample

The original proposition. In the subsection entitled "A feasible solution to linear matrix inequalities", the proposition introducing the explicit payment (source label `prop2`) asserts the following, restated in this note's notation:

> The payment t_n^{orig} in (2) satisfies the paper's sufficient matrix conditions for P1–P4, and consequently the induced mechanism has: P1, an existing and unique Nash equilibrium; P2, an equilibrium allocation solving the welfare problem; P3, zero total equilibrium payments; and P4, nonnegative equilibrium utility for every player relative to the zero outside option.

These are claims of the original paper, not assumptions of the counterexample below. Stochastic learning convergence is the separate property P5, treated subsequently in the paper.

Proposition 2. *The explicit original payment need not be individually rational, even with smooth strongly concave deterministic valuations and an efficient, budget-balanced equilibrium.*

Proof. Set $N = 3$, $K = 1$, $c = \gamma = 1$, $X_n = [0, 2]$, and

$$V_1(x) = 5x - \tfrac{1}{2}x^2, \qquad V_2(x) = V_3(x) = -x - \tfrac{1}{2}x^2.$$

These valuations are normalized and 1-strongly concave. Deterministic losses satisfy the stochastic model as a special case. Consider

$$x^* = (1, 0, 0), \qquad p^* = (4, 4, 4).$$

For every player the Hessian of its own utility in (x_n, p_n) is

$$H = \begin{pmatrix} -1 & 1/2 \\ 1/2 & -1 \end{pmatrix}, \qquad \text{eig}(H) = \{-3/2, -1/2\}.$$

Thus own utility is strictly concave. At this profile the own gradients are $(0, 0)$ for player 1 and $(-5, 0)$ for players 2 and 3. The latter players are at the lower quantity boundary. These conditions prove global best responses over $[0, 2] \times \mathbb{R}_+$, hence a Nash equilibrium. The allocation also solves (1): players 2 and 3 have strictly decreasing valuations, while player 1's valuation is increasing throughout $[0, 1]$.

Direct substitution in (2) gives

	$t^{(1)}$	$t^{(2)}$	$t^{(3)}$	$t^{(4)}$	t_n^{orig}	V_n	U_n
1	0	0	4	2/3	14/3	9/2	−1/6
2	0	0	−2	−1/3	−7/3	0	7/3
3	0	0	−2	−1/3	−7/3	0	7/3

Transfers balance, but player 1's utility is negative. It would prefer the stipulated outside option of zero. This does not contradict its best response within the game: requesting zero is still subject to the payment rule, whereas opting out is the external participation benchmark. □

Exactly what this disproves. The example directly contradicts **P4**: player 1 receives $-1/6$ at a Nash equilibrium. Thus the original proposition's combined P1–P4 conclusion is false. This counterexample does not contradict **P1**, the claim of equilibrium existence and uniqueness. It satisfies **P2** and **P3**: allocation is optimal and transfers sum to zero. It therefore does not disprove either of those properties. Nor does a static example alone disprove the separate convergence claim P5. A player can have negative utility even if a learning process converges perfectly to this equilibrium.

The claimed satisfaction of the sufficient conditions also fails directly. In the original parameterization, the linear-in-price part of the tax is $\sum_m p_m^\top a_m^n$. The paper's sufficient participation conditions include $\sum_m a_m^n \le 0$. Collecting the capacity terms in (2) instead gives

$$\sum_m a_m^n = \frac{\gamma c}{N-1} - \frac{\gamma c}{N} = \frac{\gamma c}{N(N-1)}.$$

This is positive on every positive-capacity coordinate; in the example it is $1/6$. Hence the proposition's assertion that its explicit payment satisfies the sufficient participation conditions is false. This does not refute the separate conditional participation proposition for coefficients that actually satisfy those conditions.

3 A continuum of price equilibria

Proposition 3. *The original game can have infinitely many Nash equilibria, even with a unique efficient allocation and increasing, strongly concave valuations.*

Proof. Set $N = 3$, $K = 1$, $c = 3$, $\gamma = 1$, and, for every player,

$$X_n = [0, 1], \qquad V_n(x) = 2x - \tfrac{1}{2}x^2.$$

For every $q \in [0, 1]$, consider the profile

$$x_1 = x_2 = x_3 = 1, \qquad p_1 = p_2 = p_3 = q.$$

At this profile, differentiation of the original utility gives

$$\partial_{x_n} U_n = 1 - q \ge 0, \qquad \partial_{p_n} U_n = 0.$$

Each quantity is at its upper boundary. The own utility Hessian, in (x_n, p_n), is again

$$\begin{pmatrix} -1 & 1/2 \\ 1/2 & -1 \end{pmatrix} \prec 0.$$

Thus the boundary first-order conditions suffice for global best responses, including at $q = 0$. Every displayed profile is a Nash equilibrium. All four payment terms vanish: prices agree, total demand equals capacity, and $x_n = c/N = 1$. Each player has utility $V_n(1) = 3/2$. □

This example directly disproves the uniqueness conclusion in **P1** for the explicit payment. It preserves existence, efficiency (P2), budget balance (P3), and participation (P4). All equilibria displayed have the same allocation but different price messages. Strict concavity in each player's own strategy ensures a unique best response to fixed opponents, not a unique joint equilibrium. Deleting the opponent-only term preserves this continuum, so the participation repair below does not repair full-equilibrium uniqueness. The theorem proves uniqueness only of the allocation.

4 Deleting the opponent-only adjustment

Define the repaired payment by keeping just the first three terms:

$$t_n^c(s) = t_n^{\text{orig}}(s) + \frac{\gamma}{(N-1)^2} \sum_{m \neq n} p_m^\top (x_m - c/N). \tag{3}$$

All unilateral utility differences are unchanged. Consequently the original and repaired games have exactly the same best responses and Nash equilibria; only utility levels and transfers change.

Theorem 4 (Static implementation and participation)**.** *Assume each V_n is continuously differentiable and m-strongly concave on X_n, with $V_n(0) = 0$, and $\gamma \leq m(N-1)^2$. Assume the planner has a primal-dual optimum (x^*, λ^*) satisfying its KKT conditions. Then the repaired game has an equilibrium, and every equilibrium has the planner's unique optimal allocation, common prices, exact budget balance and individual rationality. Equilibrium prices need not be unique.*

Proof. Put $S = \sum_n x_n$. The negative own price gradient is

$$F_{p_n} = \nabla_{p_n} t_n^c = \gamma \left[p_n - \frac{1}{N-1} \sum_{j \neq n} p_j - \frac{S-c}{N-1} \right].$$

At any equilibrium, orthant optimality gives $F_{p_n} \geq 0$ and $p_n^\top F_{p_n} = 0$. Summing the gradients yields $\sum_n F_{p_n} = \gamma N(c-S)/(N-1)$, hence $S \leq c$. For $\bar{p} = N^{-1} \sum_n p_n$ and $r = c - S \geq 0$, summed complementarity reads

$$0 = \frac{\gamma N}{N-1} \sum_n \|p_n - \bar{p}\|^2 + \frac{\gamma}{N-1} \left(\sum_n p_n \right)^\top r.$$

Both terms are nonnegative. Therefore $p_n = p$ for all n and $p^\top (c - S) = 0$. Allocation optimality becomes

$$\langle -\nabla V_n(x_n) + \gamma p, z_n - x_n \rangle \geq 0 \qquad (z_n \in X_n).$$

Together these are precisely the planner KKT conditions with $\lambda = \gamma p$, so every equilibrium allocation is optimal and is unique by strong concavity.

Conversely, set every price to λ^*/γ at a planner primal-dual optimum. The displayed first-order conditions hold. They suffice for global best responses because each player's negative utility has own Hessian bounded below by

$$\begin{pmatrix} mI & -\gamma I/(N-1) \\ -\gamma I/(N-1) & \gamma I \end{pmatrix} \succeq 0,$$

by $\gamma \leq m(N-1)^2$. For continuously differentiable strongly concave valuations the same conclusion follows by subtracting the strong-convexity quadratic, without requiring second derivatives. This establishes existence.

At common prices the repaired transfer simplifies to

$$t_n^c = \gamma p^\top \left(x_n - \frac{1}{N-1} \sum_{j \neq n} x_j \right),$$

so summing gives zero. Player n can deviate to $(0, p)$, whose tax is $-\gamma p^\top \sum_{j \neq n} x_j/(N-1) \leq 0$. Its utility from that deviation is nonnegative. Equilibrium optimality therefore gives $U_n^c(s^*) \geq 0$. □

In the counterexample, repaired taxes are $(4, -2, -2)$ and utilities are $(1/2, 2, 2)$. The equilibrium is unchanged, transfers balance, and all players benefit from participation. These static results do not guarantee capacity feasibility away from equilibrium, truthful type revelation, incentives to follow a prescribed learning algorithm, or uniqueness of the complete quantity–price profile.

References

[1] S. Garjani, M. Shokri and H. Kebriaei. *A Systematic Approach to Mechanism Design with Stochastic Dynamic Stability*. arXiv:2608.29130, 2026. https://arxiv.org/abs/2608.29130.